Adventure of Corgi

50.033 Game Design and Development

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- Aku Paloheimo | Sprites, Assets

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Introduction

- Our game will be a **puzzle platformer** with **pixel art**.
- The **main character** of our game is a cute **corgi named Corgi,** which should appeal to people who like the cute games.
- The Corgi in the game will **move through the stages** and the player has to ensure that it **does not die from running into enemies or traps** by **placing** platforms, walls and springs **in the right spots**.
- We would also like to add **a collectible element,** which are dog bones, to the game, to add another level of difficulty for completionists.
- The game has a few **core mechanics,** with **puzzles** being the **main one**. The other core mechanics are **management and strategy**.
- Our game differs from similar games because the player has to **collect the items** by themselves and **utilise them** to solve the puzzle.
- We will make the game for **PC** with mouse and keyboard as inputs.
- This game is **targeted** at players who **love puzzle platformers**.

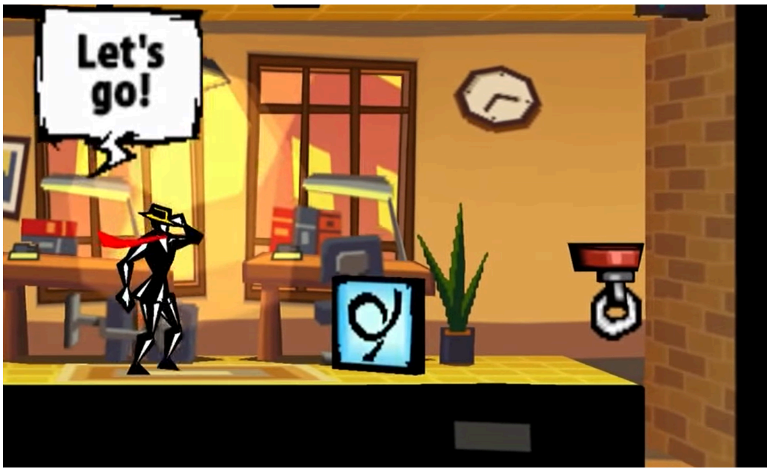
Background Study

Similar Games and Inspiration

Good Night Mr.Snoozleberg

- **Point-and-Click Puzzle Adventure**
- The Main Character constantly **moves forward without input** from the player.
- Players must **help him get through stages** by **adjusting and moving** various objects in the stage to ensure that he is **not disturbed.**
- In **Adventure of Corgi (AoC)**, Corgi will be **constantly moving in one direction**, only **changing** to the opposite direction when it **hits a wall or obstacle**.

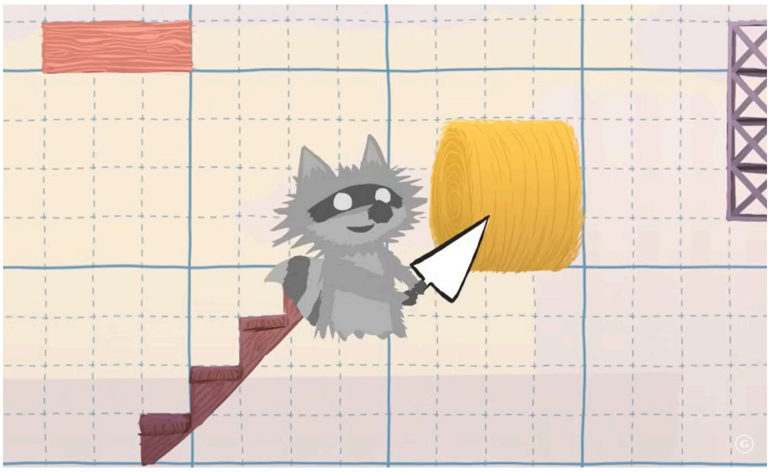
Exit



- **Puzzle platformer.**

- Player **picks up items** scattered throughout the stage and **uses them** to get through various **obstacles.**
- In **AoC**, Corgi will be able to **pick up pieces** in bubbles placed throughout to **build** on the stage.
- The player will **not have all the pieces** they need to clear the stage with right from the **start**, and need to **determine their path** through any given stage.

Ultimate Chicken Horse



- **Multiplayer Party Platformer**
- Players **create their own path** from the start to the end **using various pieces** provided at the start of each round.
- Players can place **traps**, which can be **blocked** by other players with platforms.
- In **AoC**, the player will **utilise the various pieces** they pick up to either **avoid traps or enemies**, and **create a safe path** to the end.

Motivation

Comparison to Games in Background Study

Good Night Mr Snoozleberg

- Since the player **cannot control** their character's movement, they are forced to **race against the clock**.
- Requires players to have **good timing** and be able to **find the solution** before their character reaches each obstacle.
- In **Adventure of Corgi (AoC)**, players will have the ability to **stop their character in place**, which is needed to solve various puzzles.
- (E.g., Staying on a **falling platform** to get past a **long drop** instead of **building a path down**; **waiting** for an **enemy to pass by** so that the path is safe)
- Aside from stopping Corgi, players will **not have control** over Corgi(such as making it **jump or change directions**).
- Hence players need to consider their **limited navigational abilities** when placing their pieces.

Exit

- In **Exit**, the pieces you pick up are **specifically tied to their respective obstacles.** (E.g., Fire Extinguisher on Fires, Rope for long vertical drops)
- In the **AoC**, the pieces will not be directly tied to a specific objective.
- Players will have **multiple approaches** to getting past **obstacles.** (E.g, Spring to bounce over or platform to walk over a spike pit)
- However, players need to **consider the limited number of pieces** they get within a stage when placing pieces, so **not every combination of solutions will work** to get through the entire stage.

Ultimate Chicken Horse

- Although players can place their own platforms and traps, they are **incentivised** to arrange them in a way that is **difficult for other players** to navigate.
- **Greater emphasis** on player's **platforming skills** over their ability to solve puzzles.
- **AoC** is **Single Player** and stages are **prefabricated**.
- Combined with the **lack of movement abilities**, there is **more emphasis** on the player's ability to **solve the existing puzzle** rather than their **platforming ability**.

Special Features and Uniqueness

- The special **feature** that is unique to this game is that the player can collect items in **any order** if they **can reach them,** and **utilise them however they want.**
- This game provides **more freedom** to the players for beating the stage using **various unique mechanics** of the game.
- Unlike other platformer games, the player **does not actively control the character's movement**. They can **only stop** Corgi and set the environment for the character to move accordingly.

Main Core Drives

- The main core drive that should motivate the player the most is the **empowerment of creativity & feedback.**
    - The player has to solve the puzzle **through their creativity and wisdom**.
    - Since players should collect items that they think are useful and utilise them as they want to, the player's desires to solve the stage should be **very creative in his/her unique ways**.
    - The player should feel that he/she is **smart enough to solve the stage**, and thus be **motivated to prove that he/she can also clear the next stage**.
- The second core drive of this game is **accomplishment**.
    - The player can collect **0-3 bones** while clearing each stage.
    - The player would feel **more accomplished and satisfied** as he/she gets more bones from the stage, so he/she would be **motivated to play the stage again and again** to collect more bones than before.

Targeted Player Types

- Players who prefer to play **a platformer game using their brain** rather than their platforming skills.
- Players who are eager to **prove their creativeness and smartness** by solving seemingly difficult puzzle platformer stages.
- Players who love to find their **own unique way to solve platformer puzzles**.
- Players who like **cute pixel graphics**.

The Game

Game Description

- The player is trying to **guide a corgi named Corgi through the stage to reach the end** by placing the collected items in the stage.
- Help Corgi **avoid various threats** like enemies and spikes.
- The player can **hold down "spacebar" to stop Corgi** and **place the item by clicking the screen**.
- The main storyline of this game is that our cute **corgi lost his owner,** Chaewoon, while Chaewoon was in the toilet, so begins **his adventure back to Chaewoon**.

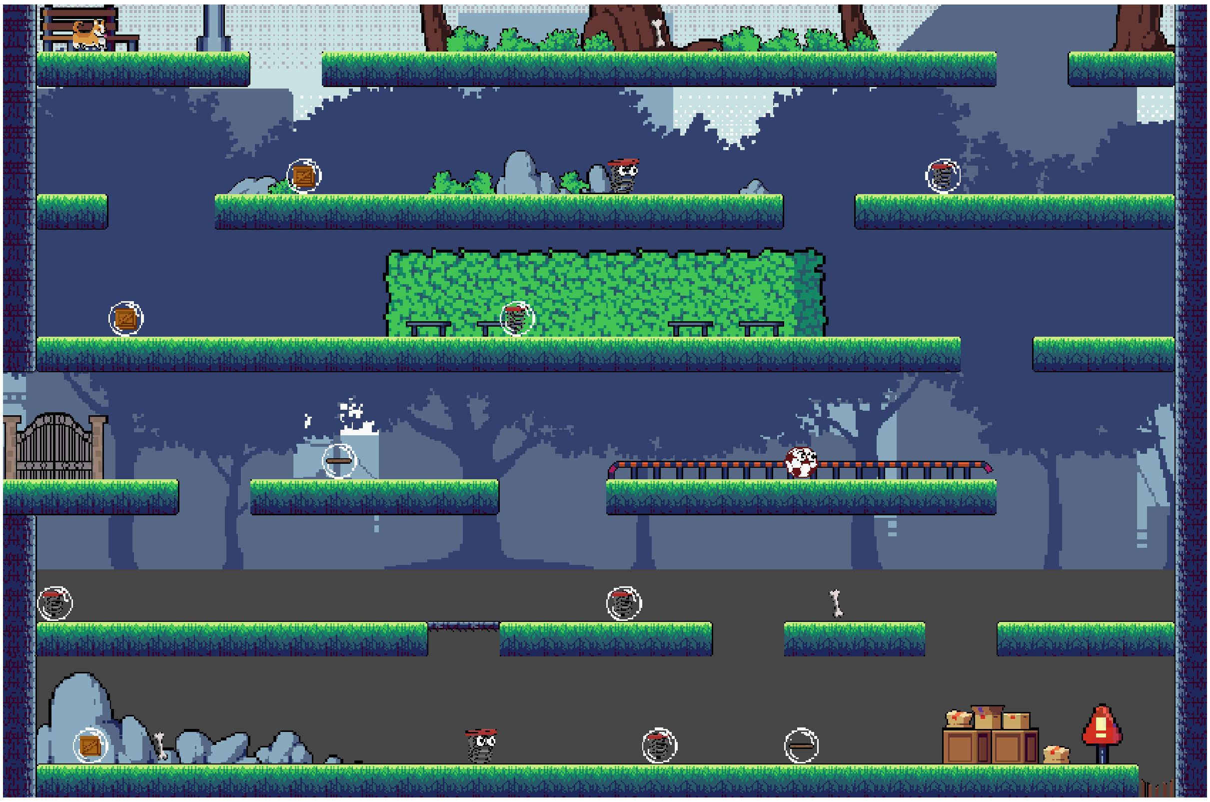
Game Core Mechanic

- The main core mechanic of this game is **puzzles**. The player would need to figure out how to guide Corgi safely towards the end of the stage. To do so, they have to analyse and solve the puzzle each level is based on.
- The other core mechanics applied is **strategy**. The player has to make long-term decisions, such as which path to take and whether to use this item now or later, to successfully reach the end and beat the stage.
- The player has **limited control** over Corgi, so **placing items** in the right places is key to clear the stage. **Management** of those collected items is, therefore, another core mechanic of this game.

Game Formal Elements

Player

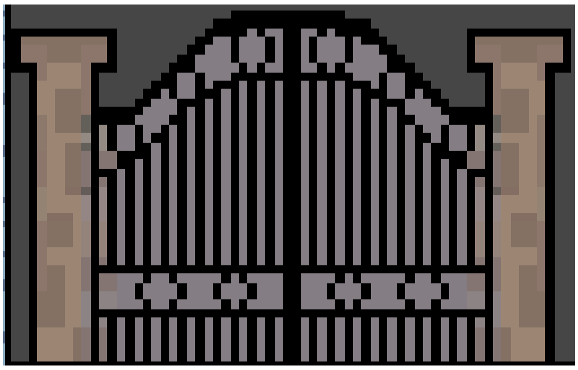
- The game is strictly a **single player** experience. The player is in charge of **guiding a corgi through several obstacles to reach the end** of the stage. In addition, Corgi can **pick up certain items** during the stage itself, which the player can then **place down** in order to **alter the stage**. Using these items in the **correct place and sequence** allows the player to **create a path** for Corgi to **reach the end goal**.



<Stage 1>

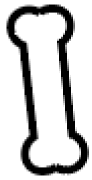
Objective, Boundaries, and Outcome

- The **main objective** is to **guide** Corgi through the stage to reach the end.



<End point of this stage>

- Two **brief tutorial stages** have been created, followed by **3 actual stages** that are meant to **test the player** on the skills they've learned.
- The **bonus objective is collecting bones** while clearing the stage. There are **three bones present in each stage**. The progress made by the player in collecting all the bones will be displayed in the **level select menu**.



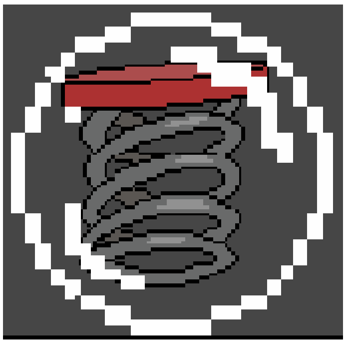
<Collective item bone>

- After **clearing the stage**, the player can **choose to proceed to the next stage, try this stage again, or return to the stage select menu** which should look like this:

<Stage select screen>

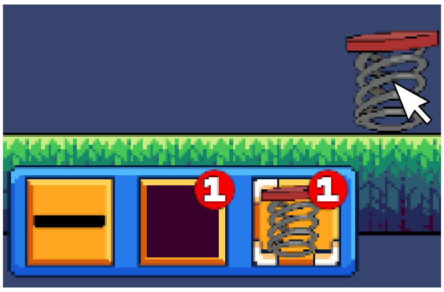
Rules and Procedures

- The player has **limited methods** to control Corgi's **movements** in the game. Corgi will **start automatically moving** to the **right side** of the screen from the beginning of all levels, regardless of whether there are any **hazards in that direction**.
- If Corgi runs into **an enemy or spikes,** the **game will be over**.
- If Corgi runs into a **block or a wall**, Corgi will **turn around** and start to **move in the opposite direction**.
- Corgi can **get the items by touching the bubbles** inside the stage. All the items collected in the stage can be **placed by the player**.
- Items are **not carried over between stages**.



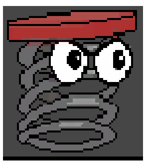
<The bubble with a spring item>

- **Item placement** will occur via **left-click of the mouse.** You can choose an item to place by **scrolling the wheel of a mouse, clicking the item from the inventory at the bottom, or pressing the appropriate number key**. The **main control scheme** is via **keyboard and mouse.**



<Example of placing items>

- The player **can place** collected items on **any visible space** in the map, however **existing items, structures or enemies** will **prevent** the player from **placing** the item in that location.
- **Enemies will roam a certain area** of the map. If Corgi **collides** with the side of the enemy, the **game will be over**.
- There also exist **falling platforms** which Corgi can stand on, triggering it to **fall after a short duration**.
- Enemies can be killed by dropping this falling platform over them.
- The player's only way to **control** Corgi is by **stopping its movements.** This can allow the player to better **think through their options** or **wait for an enemy** to move into a better position.
- In addition, different **enemies** will have **different interactions** with the **Corgi**. If Corgi were to **fall onto a Spring Enemy from above**, Corgi would **bounce upwards.** If Corgi were to **fall onto the Ball Enemy from above**, Corgi would **ride it, and jump horizontally** when the Ball Enemy reaches the edge of its platform. However, **colliding with any enemy horizontally** would lead to a **game over**.



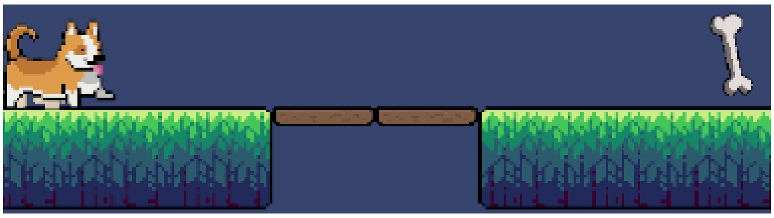
<Cute Spring Enemy>

<Cute Ball Enemy>

- The player can **restart** the stage by clicking the **restart button on the top right** of the screen.
- The game automatically **saves the progress** of the game: unlocked stages, cleared stages and number of bones collected in each stage. This allows the player to **reload his/her last status** in the game and continue.

Resources and Conflicts

- There are **three types** of items in our game that the player can place: **Normal platforms, Blocks and Springs.**
- **Normal platforms** can be used to **avoid spikes or fill out the gap** between the platforms.

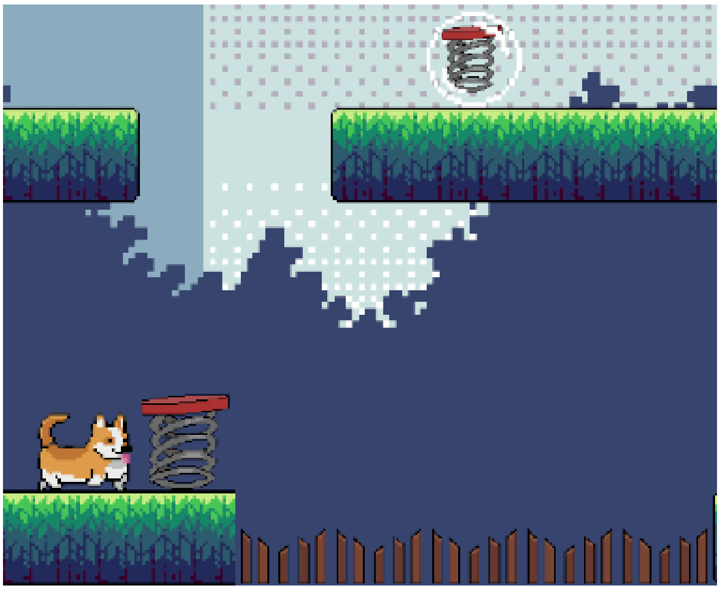


<The player placed normal platforms to fill out the gap>

- **Blocks** can be used to **change Corgi's movement direction**.

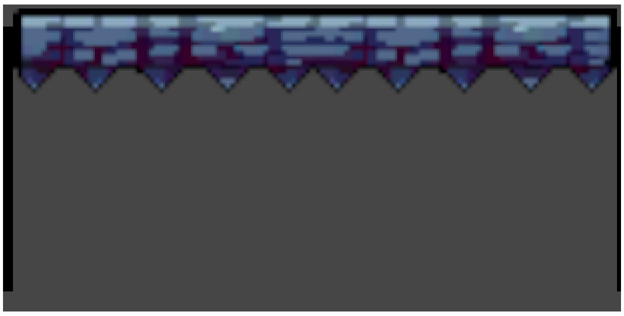
<The player place a block to change the direction of Corgi>

- **Spring items** can be used to make Corgi **bounce over to the upper floor.**



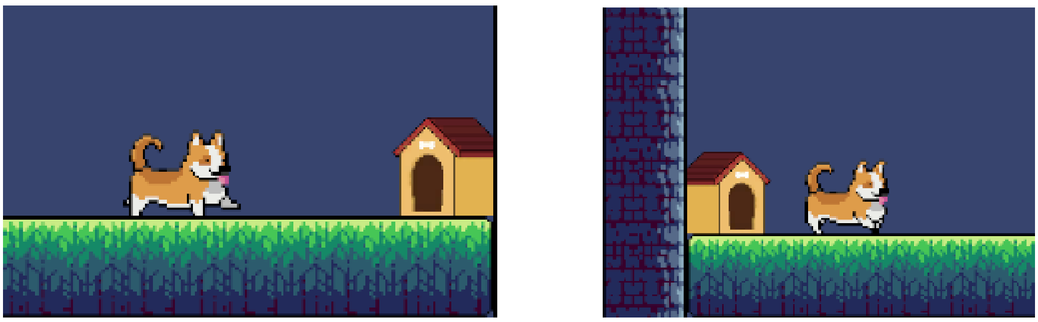
<The spring and the platform only reachable by using spring>

- When Corgi walks above a **falling platform**, the platform will **fall after a short period of time.**
- **Enemies will be eliminated** if they are **hit by a falling platform.**
- **Corgi can wait** until the **enemy moves below the falling platform** before moving onto the platform to drop it and kill the enemy.



<Falling platform>

- When Corgi **enters the dog house**, Corgi will **exit from the other dog house with the same roof colour**.



<Corgi entering and exiting from dog house>

- When Corgi **steps on the pressure plate** (red switch), it will **change the stage**. **Switch blocks** that were empty before will become **solid**, while **other switch blocks** that were solid will become **empty**.

<Red switch disabling a switch block>

- Players can increase **the difficulty of stages** by also attempting to **collect all three bones** in addition to reaching the goal.
- To **solve the puzzles**, players need to **consider the order and position of item placements.**
- E.g., Players could use two platforms to pass through a spike trap, but later on encounter another trap which also requires the use of those two platforms.

User Testing

Hypothesis

- We mainly wanted to evaluate two aspects: how easy it is to understand our game mechanics and the difficulty of the levels
- Our game is a puzzle platformer game, so we intended to make two tutorial levels which give an explanation about how the game actually works.
- We wanted to test out whether the tutorial level can introduce the in-game mechanics to the users who are new to our game.
- For the actual stages, we wanted to test the difficulty. Specifically, whether players can solve the stages, whether players can solve the puzzle in the intended way, and whether players can understand the new in-game resources.
- Of course, we also tested the overall aesthetic and bugs.

Round 1 Testing

- After we finished implementing basic item placing, Corgi movement, and two tutorial stages, we executed round1 testing to find any bugs.
- We conducted this round of testing among four group members. This excluded Seyong who implemented tutorial stages.
- Overall, we were satisfied with the difficulties of the tutorial stages. We also agreed that these two tutorial stages introduce our game mechanics and elements well and intuitively.
- Corgi movement and item placing generally worked well as intended. Some commented that Corgi movement speed was a bit slow and Corgi jump force seemed a bit weak.
- There was an unexpected bug. When we restarted the game, the enemies started moving in the wrong direction.
- After this testing round, we had several things to improve in the future. Firstly, the users said we need to add some instructions in the tutorial stages for other users who would be completely new to our game. Secondly, we needed to adjust Corgi movement speed, Corgi vertical and horizontal jump forces. We should experiment with many values to ensure that users would not feel bored and that the gameplay would not be affected because of Corgi walking too fast or jumping too high or far. Lastly, we should fix the unexpected bug from the enemy.

Round 2 Testing

- After the round 1 testing, we improved it based on the results from the round 1 testing. We experimented and re-assigned values for Corgi’s movement speed and

jump force (horizontal and vertical). We also fixed the bugs we found. In the tutorial stages, we planned to show some popups to give instructions, but we have not implemented them yet.
- We conducted this round with three users outside our group. The main objective of this testing round is to see how the outside users, who are completely new to our game, would feel with our game in tutorial stages.
- One user commented that our game has a very good original idea, so he was very interested to play it.
- Another user liked it because the gameplay was very intuitive, so he could easily understand what was going on and what he should do while playing the game.
- All three users said that we should give some instructions in the tutorial stages that explicitly mention the interaction between the different mechanics.
- They suggested to us that we should add more sprites as background elements, as without them the levels look too homogeneous and were lacking polish. Another mentioned that platform items and falling platforms looked very similar and more effort needed to be made to distinguish these two sprites.
- After completing the tutorials, one user told us to make the actual stages much more complex than the tutorials, so that they would be challenging.
- There was one more unexpected bug. If the user pressed spacebar many times before and while Corgi was falling, then Corgi changed its direction. We would fix this bug by enabling the direction change only when Corgi is in 'walk' state, not when he is in 'jump' or 'fall' state.
- After this testing round, we had several things to implement and improve. As planned, we should implement the popups for instructions in the tutorial stages. Next, we should implement actual stages, which would be very important in our puzzle game. Lastly, we should focus more on sprites, additional scenes, such as main menu, stage select, game over and stage clear, and backgrounds.

Round 3 Testing

- After Round 2, we fixed the bug that occurred, changed the sprites to make it clearer, and implemented three actual stages. We had not finished instruction popups, additional scenes and backgrounds yet.
- We conducted this round of testing with two users outside the group, who never tried our game before. The main objective of this round is to test the actual stages. We wanted to see whether our stage difficulty was appropriate, our new elements were understood easily, and if gameplay was as intended and expected.
- Both users were able to solve all three stages, after trying several times. It was hard for them to clear the stage with all three bones, as they did not spend a very long time attempting it. Overall, they both felt satisfied with the difficulty of the levels.
- As an improvement, they suggested to us to implement instruction popups in the actual stages too, so the users could better understand the new items and elements such as dog house or spring item.
- They also found some bugs. Firstly, while riding on Ball Enemy, Corgi sometimes jumped before Ball Enemy reached the end. After the testing, we figured out this bug

occurred when another Ball Enemy in the same stage reached the end. We would fix it by allowing only Ball Enemy with Corgi to invoke the Corgi jump action. Also, the Ball Enemy does not roll over the platform placed by the user and the solid switch block. We would fix this by making Ball Enemy triggered by them also.

- Since this would be our last testing, the final improvements would be done after this testing. Firstly, we would implement instruction popups, not only in tutorials but also in actual stages. Also, we would fix all the bugs mentioned. Next, we would finish all the scene transitions. Lastly, we would polish the backgrounds.

Task Distribution

Seyong Park | Level Designer

- Planned and designed mechanics. (e.g Enemy behaviour, placed item behaviour)
- Level-designed the stages (Tutorial and actual stages)
- Wrote tutorial information
- Created the stages with pre-fabbed items, platforms and background features
- Conducted Testings with external people and analysed the results

Kim Chaewoon | Programmer

- Implemented the designed mechanics for the game
- Coded Corgi's movement
- Implemented the enemy's AI and mechanics
- Implemented tile mapping to allow placing of items
- Fixed the significant amount of bugs

Ng Minzheng Joel | Programmer

- Handled game event systems
- Implemented loading and unloading scene managers
- Implemented inventory and pickup system as well as UI regarding item placing
- Implemented save system and display of unlocks in level select menu
- Implemented logic of tutorial elements

Chia Sian | Mechanics Design, Programmer

- Planned and designed enemy behaviour
- Implemented designed mechanics where possible
- Implemented stage clear and stage failure menus
- Polished the design of background for stages

Aku Paloheimo | Sprites, Assets

- Searched for the assets and soundtrack for the game
- Drew pixel art and animation of the characters and background of the game
- Created stage fail animation by making new sprites based on the purchased corgi sprite
- Made a general decision about the graphics
- Created main menu, level select menu as well as styled loading scene

Appendix

Asset Source Documentation

- Characters
    - Corgi: https://angryelk.itch.io/animated-corgi-sprite by Miniyeti and AngryElk

- Game World Objects
    - Level tiles: https://free-game-assets.itch.io/free-green-zone-tileset-pixel-art by https://free-game-assets.itch.io/
    - https://craftpix.net/freebies/free-industrial-zone-tileset-pixel-art/
    - https://craftpix.net/freebies/free-halloween-decorations-characters-and-items-pixel-art/
    - Loading screen background: https://craftpix.net/freebies/free-sky-with-clouds-background-pixel-art-set/?num=1&count=18&sq=clouds&pos=1
    - Stage Select: https://craftpix.net/freebies/free-level-map-pixel-art-assets-pack/?num=25&count=606&sq=ui%20pixel&pos=15
    - UI: https://crusenho.itch.io/complete-gui-essential-pack
    - Remaining Sprites by Aku Paloheimo
- Sound files
    - Background Music
        - Hurry Along from Pokémon Sword and Shield
        - Wedgehurst from Pokémon Sword and Shield
    - Sound Effects
        - FilmCow Recorded SFX from Filmcow https://filmcow.itch.io/filmcow-sfx
        - Nintendo-sfx from Prof. Natalie (MidTerm)
        - CasualGameSounds from Dustyroom https://assetstore.unity.com/packages/audio/sound-fx/free-casual-game-sfx-pack-54116
        - Minifantasy_Dungeon_SFX from Leohpaz https://leohpaz.itch.io/minifantasy-dungeon-sfx-pack
        - FreeSFX from Prof. Natalie (MidTerm)
        - Shapeforms-audio-effects from shapeforms https://assetstore.unity.com/packages/audio/sound-fx/shapeforms-audio-free-sound-effects-183649
- Code
    - Finite State Machine: From Prof. Natalie
    - Game Event: From Prof. Natalie